RYAN THIBAUDEAU

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SKILLS

Computer/Programming: Solid Edge, SolidWorks, NX, Catia, MATLAB, Python, ANSYS Mechanical, ANSYS Fluent, StarCCM, LabVIEW, Microsoft Suite
Additive Manufacturing: LPBF, FFF, FDM, SLA

RELEVANT EXPERIENCE

Space Dynamics Laboratory | Logan, UT ***Feb 2022 – Present***

Propulsion Engineering Assistant

- Design, simulate, build, test, and analyze 6 different hybrid rocket engines and monopropellant thrusters with thrust levels of 250 mN to 5N for satellite RPO maneuvering capability
- Source and assemble 100+ hardware, sensors, and data acquisition components for thruster test stands
- Develop LabVIEW and MATLAB code for testing and analysis of systems
- Provide propulsion design and range safety guidelines for an ESPA-sized satellite hydrazine propellant tank
- Emerging SME in additive manufacturing applications for satellite propulsion components and systems
- Wrote 100+ test procedures, test reports, and program final reports
- Present results at technical meetings
- Wrote customer proposals for $8M in funding
- Recipient of Student Scholar Award for Validation of Hydrogen Peroxide Monoprop Thruster Design

Utah State University | Logan, UT ***August 2021 - Present***

Graduate Research Assistant

- Design, simulate, build, test, and analyze a Solid Fuel Ramjet
- Design, simulate, build, test, and analyze 5 different hybrid rocket engines with thrust levels of 0.5N to 200N and propellants including GOX, N2O, H2O2, ABS, PMMA, and Nylon
- Source and assemble 100+ hardware, sensors, and data acquisition components for thruster test stands
- Develop LabVIEW and MATLAB code for testing and analysis of systems
- Written 10+ technical papers and presented research at conferences
- Awarded 1st Place in Masters Category at the 2024 AIAA International Student Conference
- Patent-pending for unique fuel blending manufacturing process.

Northrop Grumman | Magna, UT ***March 2021 – Feb 2022***

College Intern Technical

- Managed several projects at once to track assets within company and submit progress reports to manager
- Coordinated with more than a dozen other managers to facilitate movement of assets
- Lead weekly meetings to provide progress reports and prompt problem-solving discussions
- Assisted in development of testing procedures and IDIQ funding proposals for solid propellant aging studies

Blue Raven Solar | Orem, UT ***May 2019 – March 2021***

Engineering/Design Specialist

- Created 1000+ solar installation permit sets including electrical interconnection and hardware interfacing diagrams
- Managed 4 to 6 projects concurrently and collaborate with design and installation staff
- Researched causes of low performance of installed systems, identify solutions to achieve optimization resulting in $200,000+ per year in potential savings for the company and 20% average increase in KPIs
- Calculated 200+ structural engineering load analysis, providing detailed structural reports
- Provided quality assurance by auditing permit documentation for correctness and viability
- Developed training documents and standard procedures for operational optimization

EDUCATION

Utah State University, Logan UT

PhD in Aerospace Engineering — Expected 2027, GPA 4.0/4.0
Master of Science in Aerospace Engineering — December 2023, GPA 3.81/4.0

Brigham Young University, Provo UT

Bachelor of Science in Mechanical Engineering, Aerospace Emphasis — April 2021, GPA 3.51/4.0
Minor in Portuguese - Portuguese Honors Society

PUBLICATIONS

- "A Miniaturized Hydrogen Peroxide/ABS Based Hybrid Propulsion Systems for CubeSats," *SmallSat Conference*, 2022, https://digitalcommons.usu.edu/smallsat/2022/WD-S9-2022/
- "Development of a KMNO4 Catalyst-Infused Fuel Grain for H2O2 Hybrid Thruster Ignition Enhancement", *SmallSat Conference*, 2023, https://digitalcommons.usu.edu/smallsat/2023/all2023/65/
- "Development of a Potassium Permanganate Catalyst-Infused Fuel Grain for Hydrogen Peroxide Hybrid Thruster Ignition Enhancement", *AIAA SCITECH 2024 Forum*, https://arc.aiaa.org/doi/10.2514/6.2024-0139
- "Cold-Infusion of Catalytic Materials into 3-D Printed Fuels for In-Space Hybrid Propulsion Performance Enhancement", *SmallSat Conference*, 2024, https://digitalcommons.usu.edu/smallsat/2024/all2024/103/
- "Qualification of 3D-Printed Titanium Volume-Optimized Propulsion Tanks for Small Satellites", *SmallSat Conference*, 2024, https://digitalcommons.usu.edu/smallsat/2024/all2024/128/
- "Feasibility Assessment of High-Enthalpy Test Capability Using a Green-Propellant Hybrid Gas Generator", *AIAA SCITECH 2025 Forum,* https://arc.aiaa.org/doi/abs/10.2514/6.2025-1443
- "Feasibility Assessments of a Unique Additive Manufacturing Method for Functionally-Graded Solid Fuel Ramjet Propellants", *AIAA SCITECH 2026 Forum*, https://arc.aiaa.org/doi/10.2514/6.2026-1579
- "Development of a GOX/ABS Hybrid-Gas Generator System for the NASA Plume Surface Interaction (PSI) Experimental Campaign", *AIAA SCITECH 2026 Forum*, https://arc.aiaa.org/doi/10.2514/6.2026-2871
- "Simulation of Hypersonic Conditions Over a Blunt Body Using a Hybrid Rocket with Nitrogen Plume Cooling", *AIAA SCITECH 2026 Forum*, https://arc.aiaa.org/doi/10.2514/6.2026-2859
- "Enhancing Performance and Efficiency of Hybrid Rocket Thrusters for Satellite Deorbiting", *AIAA SCITECH 2026 Forum*, https://arc.aiaa.org/doi/10.2514/6.2026-1969

PATENTS

"INFUSED SOLID FUEL FOR HYBRID ROCKETS AND ORDNANCE", Pub. No.: US 2025/0282694 A1 (*Pending*)

PERSONAL PROJECTS

Rocketry Association, BYU | Provo, UT ***January 2018 – April 2021***

Hybrid Rocket Research Assistant | Provo, UT ***January 2020 – April 2021***

- Developed a 1000 Newton paraffin hybrid rocket motor with fuel additives for improved structural strength and increased regression rate
- Manufactured test structure with oxygen compatible materials and components
- Calibrated testing equipment for testing to verify improvement of fuel designs
- Wrote safety and standard operating procedure documentation to mitigate risks

GNC/Systems Engineer, High-Power Team ***August 2020 – April 2021***

- Wrote system engineering guidelines to translate customer requirements into sets of system and sub-system parameters and constraints
- Expanded on previous project to develop an autonomous altitude control system for competition rocket
- Developed 10+ designs in SolidWorks and code in MATLAB/python for controlling altitude control and rocket recovery systems

Structural Engineer, High-Power Team ***September 2019 – May 2020***

- Prepared 30+ structural SolidWorks designs, drawings, and manufacturing instructions
- Ran 20+ FEA and FMEA models to predict and reduce errors in manufacturing and rocket performance on three different projects
- Manufactured 50+ rocket parts using metals and composites to be lighter, stronger, and cost effective
- Prepared and presented 5+ preliminary and critical design reviews to board of directors

LEADERSHIP AND VOLUNTEER ACTIVITIES

American Institute of Aeronautics and Astronautics BYU Chapter | Provo, UT ***April 2018 – April 2021***

Vice President

- Organized networking events to assist 150+ students to connect with local aerospace companies
- Ran 2 showcasing events to report and demonstrate aerospace development at BYU

The Church of Jesus Christ of Latter-Day Saints | Rio de Janeiro, Brazil ***November 2014 – October 2016***

Portuguese-Speaking Service Volunteer

- Organized, planned, and delivered 50 lessons a week to enhance family connection and relationship strengths
- Supervised, motivated, and instructed 4 different groups of 16 volunteers